Andrey Urakov

Yekaterinburg | tg: @andrei_urakov | urakov18.a@gmail.com | github.com/Andrey-Good

Education

UrFU (Ural Federal University) Yekaterinburg
Bachelor's degree, Applied Artificial Intelligence (2nd year) *2024 – 2028*

Relevant Coursework

Nand2Tetris, Advanced Programming, Algorithms and Complexity Analysis, Discrete Mathematics and Mathematical Logic, Probability and Mathematical Statistics, Mathematical Analysis, Vector Analysis, Machine Learning, Databases, Non-relational Databases, Computer Architecture, Computer Networks, Data Analysis & AI

Experience

IDScan Chelyabinsk
Junior Data Annotator (Full-time / Summer contract) *July 2022 – August 2022*

- Worked offline in a closed environment with sensitive personal data that the ML team later used to train CV models.
- Performed manual annotation and transcription of images to create an OCR dataset.
- Performed data cleaning and noise filtering according to technical quality criteria.

Projects (R&D / Engineering)

Active Learning SDK | *Product Logic, Active Learning, Evaluation, Lead (Team Project)* | [Source Code] 2026

- Used AI-agent-assisted development while owning the project idea, product contract, SDK workflow, evaluation design, and acceptance criteria.
- Selected acquisition strategies for the SDK and benchmark suite, balancing uncertainty, diversity, class/group balance, and random baselines.
- Researched calibration and Temperature Scaling; translated useful findings into benchmark criteria with ECE/NLL/Brier metrics and calibration-aware stop checks.
- Led evaluation and evidence work on capped real text-classification datasets: quality gate PASS; Banking77 adaptive acquisition matched capped full-train macro-F1 at 400 labels.

Local-First AI Agent Hub | *Architecture Author, SQLite, Lead (Team of 5)* | [Arch Showcase] 2025

- Defined functional scope, roadmap, and task decomposition; owned delivery timing and quality.
- Authored the Blueprint + Spec: defined the architecture, business logic, and reliability acceptance criteria.
- Designed a failure-recovery mechanism: the system restores state consistency after a sudden process crash without data loss.
- Designed a task scheduler with protection against parallel-run conflicts and stuck tasks, using atomic operations and SQLite WAL.
- Designed an isolated execution environment based on uv to prevent dependency conflicts.
- Documented the Reconcile Loop pattern for trigger management in the specification: automatically bringing the system to the target state.
- Designed an event-driven architecture: separated business logic from the API.
- Designed I/O performance optimization through hybrid storage: metadata is stored in the database, while heavy artifacts are stored in the file system.
- UX: designed the UX architecture, user flow, and states; without a full UI design.

Melanoma Detection System | *ML Engineer, PyTorch, ONNX, Lead (Team of 4)* 2025

- Coordinated the full development cycle: defined functional scope and roadmap, decomposed requirements into technical tasks with acceptance criteria, and ensured delivery timing and quality. Organized the project defense (UrFU project work): 1st place among teams from two cohorts, 98/100.
- Implemented the full training cycle for CNN-based image classification.
- Applied data-quality improvement techniques: augmentation and class-imbalance mitigation.
- Optimized the model for mobile inference: ONNX conversion and quantization.
- UI/UX: authored the concept and UI design; designed interface behavior (states/errors) and user flow.

Review Summarizer Extension | *Backend & Core Logic, NLP, JS, NodeJS, Lead (Team of 4)* | [Source Code] 2024

- Coordinated the full development cycle: defined functional scope and roadmap, decomposed requirements into technical tasks with acceptance criteria, and ensured delivery timing and quality.
- Developed the full extension architecture (Manifest v3): core structure, a custom DOM parser, scripts, and binding the logic to the HTML interface.
- Implemented the server side: proxying requests to an LLM API and caching through request hashing to reduce latency and costs.

Skills

Languages: Python (Multiprocessing, AsyncIO, Typer), SQL (SQLite)
ML/Data: PyTorch, Classical ML (linear models, SVM/kernels, Bayesian methods, PCA), Model Evaluation, Data Annotation
Engineering: Git, System Design (Local-First systems, Event-driven), Architecture Patterns (IPC, Queues), Windows environment
Soft Skills: Technical Writing (RFC/Specs), Project Leadership (5 people), Research Mindset
Human Languages: Russian (Native), English (B1 - Intermediate)